

RAHUL VENKATARAMAN

Sr. Data Engineer

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PROFESSIONAL SUMMARY:

- 10+ years of experience as a Data Engineer designing, building, and optimizing large-scale data pipelines, ETL frameworks, and cloud-native data platforms across healthcare, financial services, retail, telecom, energy, and e-commerce domains.
- Designed and deployed production-grade data orchestration pipelines using **Apache Airflow, Dagster, and AWS Glue** on AWS, leveraging ECS, EventBridge, Lambda, and S3 to ingest, transform, and deliver enterprise datasets reliably at scale.
- Built real-time streaming architectures using **Apache Kafka, AWS Kinesis Data Streams, and Spark Streaming**, enabling sub-second event processing for fraud detection, IoT telemetry, and customer analytics platforms.
- Expert in **Apache Spark (PySpark/Scala)** for large-scale distributed data processing, implementing advanced optimizations including partition tuning, broadcast joins, caching, and dynamic resource allocation on AWS EMR and Databricks.
- Proficient in cloud data warehousing with **Amazon Redshift, Snowflake, Google BigQuery, and Azure Synapse**, designing star/snowflake schemas, optimizing distribution keys, and implementing data vault modeling for enterprise analytics.
- Strong expertise in **AWS data services** including Glue, EMR, Kinesis, Lake Formation, Athena, S3, Lambda, Step Functions, DMS, and DataSync for end-to-end data lake and lakehouse architectures.
- Implemented data quality frameworks using **Great Expectations, AWS Deequ, and custom Python validation libraries**, enforcing schema validation, null checks, referential integrity, and anomaly detection across production data pipelines.
- Experienced with **dbt (data build tool)** for SQL-based transformation layers, implementing modular data models, incremental strategies, source freshness checks, and documentation for analytics engineering workflows.
- Built observability and monitoring frameworks using **AWS CloudWatch, Prometheus, Grafana, Datadog, and Collibra**, implementing metrics collection, distributed tracing, data lineage tracking, and alerting across data platforms.
- Hands-on experience with Agile/Scrum methodologies, leading sprint planning, architecture reviews, and mentoring junior data engineers across geographically distributed teams.

TECHNICAL SKILLS:

Programming Languages: Python, PySpark, Scala, SQL, Bash, Java, HiveQL, R

Big Data & Processing: Apache Spark, PySpark, Spark Streaming, Apache Kafka, Apache Flink, Apache Hive, HDFS, MapReduce, Yarn, Apache NiFi, Sqoop, Flume, Zookeeper

Cloud – AWS: S3, Glue, EMR, Lambda, ECS, EKS, Kinesis (Streams & Firehose), Redshift, Athena, Lake Formation, DMS, DataSync, Step Functions, EventBridge, SQS, SNS, CloudWatch, CDK, CloudFormation, IAM, VPC, Secrets Manager, SageMaker

Cloud – Azure & GCP: Azure Data Factory, Azure Databricks, Azure Synapse, Azure Data Lake Storage Gen2, Azure Event Hubs, Azure SQL, CosmosDB, GCP BigQuery, GCP Dataproc, GCP Pub/Sub, GCP Cloud Storage, GCP Dataflow

Data Warehousing & BI: Amazon Redshift, Snowflake, Google BigQuery, Azure Synapse, Teradata, Star/Snowflake Schema, Data Vault, Tableau, Power BI, Amazon QuickSight, Looker, Splunk

Databases: PostgreSQL, MySQL, Oracle 12c/11g, MS SQL Server, MongoDB, Cassandra, DynamoDB, Redis, Elasticsearch, HBase

ETL & Orchestration: AWS Glue, Apache Airflow, Dagster, dbt, Informatica PowerCenter 10.x, Talend 6.x, SSIS, AWS Step Functions, Prefect

Data Quality & Governance: Great Expectations, AWS Deequ, Apache Atlas, Collibra, Alation, ERWIN, AWS Lake Formation

DevOps & CI/CD: Jenkins, Docker, Kubernetes, Helm, Terraform, AWS CDK, Git, GitHub, GitLab, SonarQube, Maven, Gradle

Monitoring & Observability: AWS CloudWatch, Prometheus, Grafana, Datadog, Splunk, ELK Stack, PagerDuty, AWS X-Ray

ML & AI Integration: AWS SageMaker, MLflow, DSPy, LangChain, NumPy, Pandas, Scikit-Learn, MLLIB

Methodologies: Agile/Scrum, SAFe, Data Mesh, Data Lakehouse, Lambda/Kappa Architecture, TDD, SDLC

PROFESSIONAL EXPERIENCE:

UnitedHealth Group (Client: Optum Analytics): October 2023 – Current (Location: Eden Prairie, MN (Remote))

Role: Sr. Data Engineer

Responsibilities:

- Designed and deployed HIPAA-compliant **AWS Glue and Apache Spark ETL pipelines** ingesting 50M+ daily healthcare claims, member eligibility records, and clinical encounter data from HL7 FHIR APIs and EDI 837/835 feeds into an AWS S3 data lake partitioned by payer, date, and claim type.
- Built real-time streaming pipelines using **AWS Kinesis Data Streams and Apache Kafka**, processing prior authorization events and pharmacy benefit transactions with sub-500ms latency, triggering downstream Lambda enrichment functions and Redshift Streaming Ingestion for live dashboards.
- Implemented **dbt transformation models on Amazon Redshift** building a semantic layer for HEDIS quality measures, risk adjustment factor (RAF) scoring, and population health KPIs, enabling self-service analytics for 200+ clinical analysts via Tableau and QuickSight.
- Orchestrated end-to-end pipeline workflows using **Apache Airflow on AWS MWAA**, managing DAG dependencies for claims adjudication, provider network updates, and member enrollment backfills, implementing SLA alerts and retry policies with CloudWatch and PagerDuty integration.
- Built a **AWS Lake Formation governed data lakehouse** with fine-grained column-level access controls for PHI data, implementing AWS Glue Data Catalog, row-level security policies, and HIPAA-compliant audit trails via CloudTrail and Collibra lineage tracking.
- Developed **Great Expectations data quality suites** validating referential integrity between member IDs, provider NPIs, and claim line items, embedding checkpoint validations in Airflow DAGs and publishing HTML quality reports to S3 for compliance review.
- Engineered **AWS CDK infrastructure-as-code** provisioning EMR clusters, Glue job environments, Kinesis shards, Redshift clusters, and Lake Formation permissions, enabling repeatable environment deployments with HIPAA/HITECH compliance guardrails.
- Integrated **AWS SageMaker ML pipelines** with data engineering workflows, building feature stores and training dataset preparation pipelines in PySpark for readmission risk, chronic disease progression, and pharmacy adherence prediction models.
- Implemented **Collibra data governance workflows** for PHI classification, data stewardship, business glossary management, and automated impact analysis across 300+ datasets in the enterprise data lake, ensuring HIPAA, HITECH, and CMS compliance.

- Built **AWS Step Functions state machines** orchestrating multi-step ETL workflows for CMS regulatory reporting (HEDIS, STARS, ACA MLR), coordinating Glue jobs, Lambda validators, Redshift loads, and SNS notifications with error compensation logic.
- Optimized **PySpark jobs on AWS EMR** processing historical claims backfills of 5+ billion records, implementing adaptive query execution, dynamic partition pruning, and columnar Parquet storage reducing job runtime from 14 hours to 2.5 hours.

Environment: *Python, PySpark, Apache Spark, AWS Glue, AWS EMR, AWS Kinesis, Apache Kafka, Amazon Redshift, dbt, Apache Airflow (MWAA), AWS Lake Formation, AWS Step Functions, AWS CDK, Great Expectations, Collibra, AWS SageMaker, MLflow, Tableau, Power BI, CloudWatch, PagerDuty, Terraform*

JPMorgan Chase & Co.: June 2021 – September 2023 (Location: Columbus, OH)

Role: Sr. Data Engineer

Responsibilities:

- Built PCI-DSS and SOX-compliant **Apache Kafka event streaming pipelines** ingesting real-time payment card transactions, wire transfers, and ACH events from 40+ source systems, processing 8M+ daily events into S3 data lake and Redshift for fraud analytics and regulatory capital reporting.
- Designed **Snowflake data warehouse** on AWS with multi-cluster configurations, implementing data vault 2.0 modeling for customer 360, account hierarchy, and transaction history, creating role-based access controls for 1,500+ analytics consumers across risk, compliance, and business intelligence teams.
- Developed **PySpark Spark Structured Streaming jobs on AWS EMR** for real-time fraud scoring, joining card transaction streams with merchant risk profiles and behavioral models from DynamoDB, triggering Lambda-based case creation in the fraud operations platform within 150ms.
- Implemented **AWS DMS and custom Python CDC pipelines** migrating 200TB of Oracle and DB2 mainframe banking data to S3 data lake, designing incremental extraction strategies using log-based CDC, watermark columns, and Glue bookmarks for zero-downtime migration.
- Built **dbt projects on Snowflake** modeling regulatory datasets for Basel III RWA calculation, CCAR stress testing, and FRTB market risk, implementing incremental materialization strategies and automated documentation with dbt docs deployed on S3.
- Orchestrated complex financial data workflows using **Dagster on AWS ECS**, managing asset-based pipelines for trade settlement, position reconciliation, and P&L reporting, implementing sensor-based triggers for real-time market data events from Bloomberg and Reuters feeds.
- Engineered **AWS Glue DataBrew transformation recipes** standardizing customer PII data, applying tokenization, format normalization, and de-duplication across 80+ source feeds from retail banking, investment banking, and credit card platforms under GLBA compliance.
- Implemented **Apache Atlas and AWS Glue Data Catalog** for enterprise data lineage, classifying 500+ datasets with business metadata, PII tags, and retention policies, enabling automated data discovery and impact analysis for regulatory audit readiness.
- Built **AWS Lambda and EventBridge-driven micro-batch pipelines** for intraday position updates and trade confirmations, processing FIX protocol messages from OMS systems and loading enriched trade records into Redshift and DynamoDB with SOX-compliant audit logging.
- Developed **Terraform and AWS CDK modules** standardizing data platform infrastructure provisioning across 12 data engineering teams, including EMR cluster templates, Glue job configurations, Kinesis shard calculators, and Redshift WLM policies.

Environment: *Python, PySpark, Apache Spark, Apache Kafka, AWS Glue, AWS EMR, AWS DMS, Amazon Redshift, Snowflake, dbt, Dagster, AWS Lambda, AWS EventBridge, DynamoDB, Apache Atlas, AWS CDK, Terraform, Jenkins, Docker, Kubernetes, CloudWatch, Datadog, Power BI*

Walmart Global Tech (Client: Sam's Club): August 2019 – May 2021 (Location: Bentonville, AR)

Role: Data Engineer

Responsibilities:

- Designed **Apache Spark batch and streaming data pipelines** on AWS EMR processing 100TB+ daily retail transaction data from 600+ Sam's Club locations, ingesting POS sales, inventory movements, membership events, and supplier feeds into a centralized S3 data lake with Delta Lake ACID transaction support.
- Built **Apache Kafka Streams and AWS Kinesis Firehose pipelines** for real-time inventory visibility, processing item scan events, RFID tags, and warehouse management system updates, enabling same-day inventory accuracy improvements from 78% to 97% across distribution centers.
- Developed **PySpark ETL jobs on AWS Glue** transforming supplier EDI 856/810 feeds, product catalog updates, and demand signals into curated data marts in Redshift, powering QuickSight dashboards for category managers, supply chain planners, and merchant teams.
- Implemented **Apache Airflow DAGs on Amazon MWAA** orchestrating end-to-end merchandise planning workflows, coordinating Spark jobs, Glue crawlers, Redshift COPY commands, and dbt model runs with SLA monitoring and automatic failure alerting via SNS and Slack.
- Built a **self-service data lakehouse using AWS Lake Formation and Athena**, enabling 300+ business analysts to query curated sales, membership, and inventory datasets using SQL without infrastructure management, reducing ad-hoc data request turnaround from 5 days to 2 hours.
- Designed **dbt models on Redshift** building member lifetime value, basket analysis, and item affinity datasets for the personalization platform, implementing incremental models with snapshot strategies for slowly changing member demographics and purchase history.
- Integrated **AWS SageMaker feature stores** with data engineering pipelines, building automated feature pipelines extracting member behavioral signals from clickstream data (AWS Kinesis + EMR) and transactional history for product recommendation ML models.
- Implemented **Great Expectations checkpoints** in Airflow DAGs validating daily sales, inventory, and membership datasets, publishing data quality scorecards to Confluence and triggering PagerDuty alerts for SLA-breaching quality failures before data reaches downstream consumers.
- Built **AWS CDK data platform templates** standardizing EMR cluster provisioning, S3 bucket lifecycle policies, Lake Formation permissions, and Glue job configurations across 8 data engineering squads, reducing new pipeline setup time from 3 weeks to 2 days.

Environment: *Python, PySpark, Apache Spark, Apache Kafka, AWS Glue, AWS EMR, Amazon MWAA, AWS Kinesis, Delta Lake, Amazon Redshift, dbt, AWS Lake Formation, Amazon Athena, AWS SageMaker, Great Expectations, AWS CDK, Jenkins, Docker, Kubernetes, QuickSight, CloudWatch, Grafana*

AT&T Inc.: March 2017 – July 2019 (Location: Dallas, TX)

Role: Data Engineer

Responsibilities:

- Built **Apache Kafka and Spark Streaming pipelines** ingesting 2B+ daily network CDR (Call Detail Records), data usage events, and IoT device telemetry from 5G and LTE network probes into AWS S3, enabling real-time network quality analytics and customer experience monitoring for 230M+ subscribers.
- Developed **PySpark data transformation jobs on AWS EMR**, processing subscriber churn prediction features, network congestion signals, and revenue assurance datasets for the enterprise analytics platform, optimizing job performance with dynamic allocation and partition coalescing.
- Implemented **AWS Glue crawler and Data Catalog** automating schema discovery across 400+ S3 data sources including network logs, billing records, and CRM exports, enabling Athena-based ad-hoc querying and Lake Formation governed access for network operations and marketing analytics teams.

- Built **Teradata to Amazon Redshift migration pipelines** using AWS DMS and custom Python extraction scripts, migrating 150TB of subscriber billing, plan, and usage data with table re-partitioning, sort key optimization, and WLM queue configuration for query performance parity.
- Designed **Apache Airflow DAGs** orchestrating telecom regulatory reporting workflows (FCC, CPUC), coordinating monthly CDR aggregation, quality validation, and encrypted file delivery to regulatory agencies with audit trail logging in S3 and Redshift.
- Developed **Python and Scala Spark jobs** for subscriber segmentation, usage pattern analysis, and ARPU (Average Revenue Per User) modeling, loading results into Snowflake for consumption by CRM, marketing automation, and product pricing teams via Tableau dashboards.
- Implemented **AWS Lambda and SQS-based event processing** for real-time provisioning of network services, processing order management events and triggering network configuration APIs for 5G service activation, eSIM provisioning, and device upgrade workflows.
- Built **data quality monitoring using AWS Deequ** on Spark, validating subscriber PII completeness, CDR record counts, and billing cycle integrity across 300+ daily data feeds, publishing metrics to CloudWatch and triggering automated data incident tickets in ServiceNow.

Environment: *Python, PySpark, Scala, Apache Kafka, Apache Spark, AWS Glue, AWS EMR, AWS DMS, Amazon Redshift, Snowflake, Apache Airflow, AWS Kinesis, Amazon Athena, AWS Lambda, AWS SQS, AWS Deequ, AWS CDK, Terraform, Jenkins, Docker, Tableau, CloudWatch, Splunk*

ExxonMobil Corporation: June 2015 – February 2017 (Location: Houston, TX)

Role: Data Engineer / ETL Developer

Responsibilities:

- Built **Apache Spark and Hadoop MapReduce pipelines** on AWS EMR processing seismic survey data, well production logs, and refinery sensor telemetry from 500+ upstream and downstream assets, ingesting 20TB+ daily operational data into S3 for reservoir modeling and production optimization analytics.
- Developed **Informatica PowerCenter ETL workflows** integrating SAP PM/ERP, OSIsoft PI historian, and SCADA systems for equipment maintenance, production KPIs, and energy consumption data, loading into Teradata enterprise data warehouse for HSE reporting and executive dashboards.
- Designed **AWS Glue Python Shell and PySpark jobs** migrating legacy Informatica pipelines to cloud-native ETL on S3, transforming drilling operations, completion costs, and hydrocarbon production data with schema evolution support and backfill automation.
- Implemented **Apache Airflow orchestration** for daily production reporting workflows, coordinating data extraction from field historian systems, Spark transformation jobs, Redshift loads, and automated Power BI dataset refreshes for upstream production teams.
- Built **real-time IoT sensor data pipelines** using AWS Kinesis Data Streams and Lambda, ingesting pressure, temperature, and flow rate readings from 10,000+ wellhead sensors, applying anomaly detection algorithms to trigger automated alerts for equipment failure prevention.
- Developed **Python-based data quality frameworks** validating production volume reconciliation between field measurement systems and accounting records, implementing automated variance reporting and exception workflows integrated with SAP and the Redshift analytics layer.
- Designed **Snowflake data warehouse schemas** for upstream exploration and downstream refining analytics, implementing multi-dimensional models for crude oil pricing, feedstock optimization, and margin analysis, integrating with Bloomberg commodity data feeds via AWS Lambda.

Environment: *Python, PySpark, Apache Spark, Hadoop, AWS Glue, AWS EMR, AWS Kinesis, Amazon Redshift, Snowflake, Teradata, Informatica PowerCenter, Apache Airflow, AWS Lambda, AWS S3, Power BI, Tableau, Grafana, CloudWatch, Jenkins, Docker*

Infosys Limited (Client: Amazon Retail): August 2013 – May 2015 (Location: Seattle, WA)

Role: ETL / Data Engineer

Responsibilities:

- Built **Python and Java ETL pipelines** integrating Amazon marketplace seller data, inventory feeds, and product catalog updates from 50+ global fulfillment centers into Amazon Redshift, enabling seller performance analytics and inventory forecasting for the Amazon Seller Central platform.
- Developed **Apache Hive and Pig scripts on Hadoop** processing clickstream logs, search query data, and customer review datasets (10TB+ daily) for product recommendation and search relevance improvement initiatives on Amazon.com.
- Implemented **AWS Glue and S3-based data lake foundations**, designing S3 partition strategies, Glue Data Catalog schemas, and Athena query optimizations for retail analytics, A/B testing result datasets, and customer behavioral cohort analysis.
- Built **Talend ETL jobs** integrating third-party logistics providers, customs data, and import/export compliance records into the supply chain analytics platform, automating daily data reconciliation and exception reporting workflows.
- Developed **Apache Sqoop and custom JDBC extraction scripts** migrating Oracle ERP and MySQL operational databases to S3 data lake, implementing incremental extraction strategies for order management, returns processing, and vendor payment datasets.
- Implemented **Apache Oozie and later Apache Airflow DAGs** orchestrating Hive, Pig, and Spark batch jobs for nightly sales reporting, customer segmentation updates, and inventory position refreshes, managing job dependencies across 200+ workflow tasks.
- Built **data quality validation scripts in Python** enforcing referential integrity between order, customer, and product datasets, generating automated data quality scorecards and escalating failures to data stewardship teams via SNS and JIRA integration.

Environment: *Python, Java, Apache Hive, Apache Pig, Apache Spark, Hadoop HDFS, Apache Sqoop, Apache Oozie, Apache Airflow, AWS Glue, Amazon S3, Amazon Redshift, Amazon Athena, Talend, Oracle, MySQL, AWS Lambda, Jenkins, Git, Power BI, Tableau*

CERTIFICATIONS

AWS Certified Data Engineer – Associate – Amazon Web Services (AWS) | Issued: January 2024 | Credential ID: AWS-DEA-2024-RV9921

AWS Certified Solutions Architect – Associate – Amazon Web Services (AWS) | Issued: March 2022 | Credential ID: AWS-SAA-2022-RV7734

AWS Certified Cloud Practitioner – Amazon Web Services (AWS) | Issued: September 2020 | Credential ID: AWS-CCP-2020-RV3312

Databricks Certified Associate Developer for Apache Spark 3.0 – Databricks | Issued: July 2023 | Credential ID: DBR-SPARK3-2023-RV0041

Databricks Certified Data Engineer Associate – Databricks | Issued: November 2023 | Credential ID: DBR-DEA-2023-RV0088

Snowflake SnowPro Core Certification – Snowflake Inc. | Issued: April 2022 | Credential ID: SNOW-CORE-2022-RV5521

Google Professional Data Engineer – Google Cloud (GCP) | Issued: August 2021 | Credential ID: GCP-PDE-2021-RV6644

Confluent Certified Developer for Apache Kafka (CCDAK) – Confluent Inc. | Issued: February 2023 | Credential ID: CCDAK-2023-CF-RV7712

HashiCorp Certified: Terraform Associate – HashiCorp | Issued: October 2022 | Credential ID: HCTA-2022-RV-00456

dbt Analytics Engineering Certification – dbt Labs | Issued: June 2023 | Credential ID: DBT-AEC-2023-RV2209

Microsoft Certified: Azure Data Engineer Associate (DP-203) – Microsoft | Issued: December 2021 | Credential ID: DP203-2021-MS-RV8812

Certified Kubernetes Application Developer (CKAD) – Cloud Native Computing Foundation (CNCF) | Issued: May 2022 | Credential ID: CKAD-2022-RV-00789

EDUCATION:

Bachelor of Technology in Computer Science & Engineering – Osmania University, Hyderabad, India, 2009–2013